

API Specification & Service Contract

Product Name: HQ

Tagline: Your AI Headquarters

Version: 1.0

Purpose

This document defines the communication contract between the frontend, backend, mobile application, AI orchestration engine, and third-party services.

The API follows a REST-first approach for the MVP, with WebSockets used for real-time updates. The architecture is designed to evolve toward GraphQL or hybrid APIs if future requirements demand it.

API Principles

Every API must be:

- Consistent
- Predictable
- Versioned
- Secure
- Documented
- Testable
- Scalable
- AI-friendly

Base URL

/api/v1

Future versions:

/api/v2

/api/v3

Authentication Module

Authentication Endpoints

- POST "/auth/register"
- POST "/auth/login"
- POST "/auth/google"
- POST "/auth/logout"
- POST "/auth/refresh"
- POST "/auth/forgot-password"
- POST "/auth/reset-password"
- POST "/auth/verify-email"

User Module

- GET "/users/me"

- PATCH "/users/me"
- PATCH "/users/preferences"
- DELETE "/users/me"

Company Module

- POST "/companies"
- GET "/companies"
- GET "/companies/:id"
- PATCH "/companies/:id"
- DELETE "/companies/:id"

Brand Module

- POST "/brands"
- GET "/brands"
- PATCH "/brands/:id"
- DELETE "/brands/:id"

Brand configuration includes:

- Logo
- Colors
- Fonts
- Voice
- Target Audience

- Products
- Social Accounts

Headquarters Module

- POST "/headquarters"
- GET "/headquarters"
- GET "/headquarters/:id"
- PATCH "/headquarters/:id"

Returns:

- Executive Board
- Departments
- Active Missions
- Statistics
- Dashboard Summary

Executive Module

- GET "/executives"
- GET "/executives/:id"
- GET "/executives/:id/status"
- GET "/executives/:id/memory"
- PATCH "/executives/:id"

Department Module

- GET "/departments"
- GET "/departments/:id"
- GET "/departments/:id/executives"

Mission Module

The heart of HQ.

Endpoints

- POST "/missions"
- GET "/missions"
- GET "/missions/:id"
- PATCH "/missions/:id"
- DELETE "/missions/:id"

Mission actions:

- POST "/missions/:id/start"
- POST "/missions/:id/pause"
- POST "/missions/:id/resume"
- POST "/missions/:id/cancel"
- POST "/missions/:id/approve"

Task Module

- GET "/missions/:id/tasks"
- GET "/tasks/:id"
- PATCH "/tasks/:id"
- POST "/tasks/:id/complete"

AI Orchestration Module

Internal APIs used by the Executive Organization.

Examples:

- POST "/ai/ceo/analyze"
- POST "/ai/chief-of-staff/plan"
- POST "/ai/executives/assign"
- POST "/ai/executives/review"
- POST "/ai/executives/approve"

These endpoints are internal and not exposed to end users.

Content Generation Module

- POST "/content/generate"
- POST "/content/rewrite"
- POST "/content/summarize"

- POST "/content/translate"
- GET "/content/history"

Image Generation Module

- POST "/images/generate"
- GET "/images"
- DELETE "/images/:id"

Asset Module

- POST "/assets/upload"
- GET "/assets"
- GET "/assets/:id"
- DELETE "/assets/:id"

Supports:

- Images
- Videos
- PDFs
- Documents
- Audio
- Brand files

Knowledge & Memory Module

- GET "/knowledge"
- POST "/knowledge"
- PATCH "/knowledge/:id"

Executive Memory:

- GET "/memory/executives/:id"
- PATCH "/memory/executives/:id"

Notification Module

- GET "/notifications"
- PATCH "/notifications/read"
- DELETE "/notifications/:id"

Analytics Module

- GET "/analytics/dashboard"
- GET "/analytics/missions"
- GET "/analytics/executives"
- GET "/analytics/content"

Billing Module

- GET "/subscription"

- POST "/subscription/checkout"
- POST "/subscription/cancel"
- GET "/payments"
- GET "/invoices"

Settings Module

- GET "/settings"
- PATCH "/settings/profile"
- PATCH "/settings/security"
- PATCH "/settings/preferences"
- PATCH "/settings/ai"

WebSocket Events

Real-time events include:

- Mission Created
- Mission Updated
- Executive Started
- Executive Finished
- Task Assigned
- Task Completed
- CEO Approved Mission
- Notification Created
- Payment Completed

These events keep the dashboard synchronized without requiring manual refreshes.

Standard API Response

Every successful response follows a consistent structure.

```
{  
  "success": true,  
  "message": "Mission created successfully",  
  "data": {},  
  "meta": {}  
}
```

Standard Error Response

```
{  
  "success": false,  
  "message": "Validation failed",  
  "errors": []  
}
```

Security Standards

Every endpoint must implement:

- JWT Authentication
- Refresh Tokens
- Role-Based Access Control (RBAC)
- Request Validation
- Rate Limiting
- Audit Logging

Sensitive operations require additional authorization checks.

API Documentation

The backend automatically generates API documentation using OpenAPI (Swagger).

Documentation includes:

Request examples

Response examples

Authentication requirements

Error codes

Rate limits

This ensures developers and AI coding assistants always work from the same API contract.

Conclusion

The HQ API is designed to provide a stable, scalable communication layer between every client application, backend module, and AI executive service.

A consistent API contract allows independent development of the frontend, mobile application, backend services, and AI orchestration engine while maintaining compatibility.

